

Bladder Augmentation and Catheterizable Channels

WARWIKI

Patient Information · A bigger, safer bladder and a private way to empty it

Bladder augmentation makes a small, stiff bladder **bigger and softer** with a patch of your own bowel. A catheterizable channel (Mitrofanoff or Monti) is a narrow tube from the bladder to a **small opening on your belly** for emptying with a catheter. They are often done together, and both need **care for life**.

About These Operations

- **Augmentation (enterocystoplasty).** A short piece of bowel, with its blood supply, is opened flat and sewn onto the bladder like a patch. The bowel is joined back together. The bladder then holds more urine at a **lower pressure**, which protects the kidneys and reduces leaking.
- **Channel.** It is made from the appendix (Mitrofanoff) or a short piece of bowel shaped into a narrow tube (Monti). One end is tunneled into the bladder wall to make a valve; the other becomes a small opening (a **stoma**) in the belly button or lower belly. It stays closed between uses, so **no bag** is needed.

They are for adults whose bladder stays small or high-pressure despite medicines and bladder Botox, often from spina bifida, spinal cord injury or another nerve condition. A channel helps when catheterizing through the urethra is hard.

Benefits and Risks in Numbers

about
3 in 10

formed bladder stones within 10 years

about
1 in 10

had a bladder tear, an emergency, within 10 years

about
1 in 2

channels needed a repair operation over about 12 years

Stone and tear numbers come from a large study of people with spina bifida. Your own risk may differ; ask your team.

Living With It

- **Catheterize on schedule, every time.** Most people need a catheter to empty. An overfull augmented bladder can tear.
- **Rinse out mucus.** The bowel patch makes mucus for life. Rinsing (irrigation) as your team teaches keeps catheters flowing and helps prevent stones.
- **Yearly checks for life:** a visit, blood tests and a kidney scan. Vitamin B12 is checked over time.
- **Cancer in the bladder is uncommon** but can appear many years later, usually after 10 years. Blood in the urine needs a camera check.

Learn the Terms

Augmentation (enterocystoplasty)

Enlarging the bladder with a patch of bowel.

Mitrofanoff channel

A catheter channel made from the appendix.

Monti channel

A catheter channel made from a short piece of small bowel.

Stoma

The small opening on the belly where the catheter goes in.

Catheterize

To pass a thin tube into the bladder to empty it.

Irrigation

Rinsing the bladder with salt water to clear mucus.

Mucus

Slippery fluid the bowel patch keeps making.

Perforation

A tear in the bladder wall. It needs emergency care.

Urodynamics

A test of bladder filling and pressure, done before surgery.

Is this a big operation? Yes. It is major belly surgery, and you are asleep for it. Most people stay in the hospital for 5 days to 2 weeks. It changes daily life for good, so take time to decide, and ask about other choices such as a urostomy bag.

In the hospital Tubes drain the bladder. You learn to flush them if mucus blocks them.	First 2 to 4 weeks The tubes stay in while you heal. An X-ray may check healing first.	Tubes out You learn to catheterize the channel and rinse out mucus.	Every year A visit, blood test and kidney scan, for life.
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How to Get Ready

- You have **urodynamics**, a kidney scan and blood tests.
- You and your team plan who will catheterize and where the stoma goes, so you can reach it easily.
- Tell your team about past belly or bowel surgery, bowel problems, and any **shunt** for fluid on the brain.
- Follow your instructions about eating, drinking and which medicines to stop.

What Happens

- 1 You are asleep and get antibiotics.
- 2 A piece of bowel is opened and sewn onto the bladder. The bowel is joined back together.
- 3 If needed, a channel is made and brought to a small opening on the belly.
- 4 Tubes are left in the bladder and the channel while they heal. Never pull out the channel tube.

Problems to Know About

- The stoma can **narrow**, so the catheter is hard to pass, or it can **leak**. A small procedure or another operation can often fix it.
- If the catheter will not pass, try once more gently with a well-lubricated catheter. **Do not force it.**
- **Germs in the urine are common** and are not treated unless you feel unwell.
- The bowel patch can upset your blood's **acid balance**, which over time can weaken bones. It can also lower **vitamin B12**. Blood tests check for both.

Call the office that did your surgery if you have:

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- Blood in the urine, or new pain over the bladder
- A catheter that is harder to pass, or a stoma that leaks

Go to an emergency department if:

- Sudden belly pain or swelling, fever, or you feel very ill. Say you have an **augmented bladder**; it may have torn.
- You cannot pass a catheter or flush a blocked tube, and your bladder feels full
- You cannot reach the office and are worried

Three Things to Remember

1. Augmentation makes the bladder bigger and safer. A channel gives a private way to empty it.
2. Catheterize on schedule and rinse out mucus for life. An overfull bladder can tear.
3. Keep your yearly checks. Report blood in the urine or sudden belly pain at once.